Tier 1 IT Home Lab #1

Network Troubleshooting

Objective

Practice a structured Tier 1 network troubleshooting process by checking the computer's IP configuration, local connectivity, internet connectivity, DNS resolution, and proxy configuration.

Troubleshooting Process

1. **Check IP Configuration**
   Run **ipconfig** to identify the IPv4 address, subnet mask, and default gateway before testing connectivity.
2. **Test the Default Gateway**
   Ping the default gateway to verify communication with the local network.
3. **Test External Connectivity**
   Ping **8.8.8.8** to verify internet connectivity without relying on DNS.
4. **Test DNS Resolution**
   Ping **google.com** and use **nslookup** to verify that domain-name resolution is working.
5. **Check Proxy Configuration**
   Review Windows Proxy settings to determine whether a manual proxy server is

configured. ### Step 1 — IP Configuration

The **ipconfig** command was used first to review the computer's network configuration and identify the information needed for the connectivity tests, including the IPv4 address, subnet mask, and default gateway.

Lab result: The IP configuration information was reviewed and used to identify the default gateway for the next troubleshooting step.

Step 2 — Default Gateway Test

The default gateway was successfully reached with **0% packet loss**, confirming local network communication.

Step 3 — External Connectivity Test

The computer successfully reached **8.8.8.8** with **0% packet loss**, confirming external connectivity.

Step 4 — DNS Testing

The computer successfully reached **google.com**, and **nslookup google.com** returned IP addresses, confirming DNS resolution was functioning.

Step 5 — Proxy Configuration Check

Windows Proxy settings were reviewed. **Use a proxy server was Off**, confirming that no manual proxy server was configured at the time of testing.

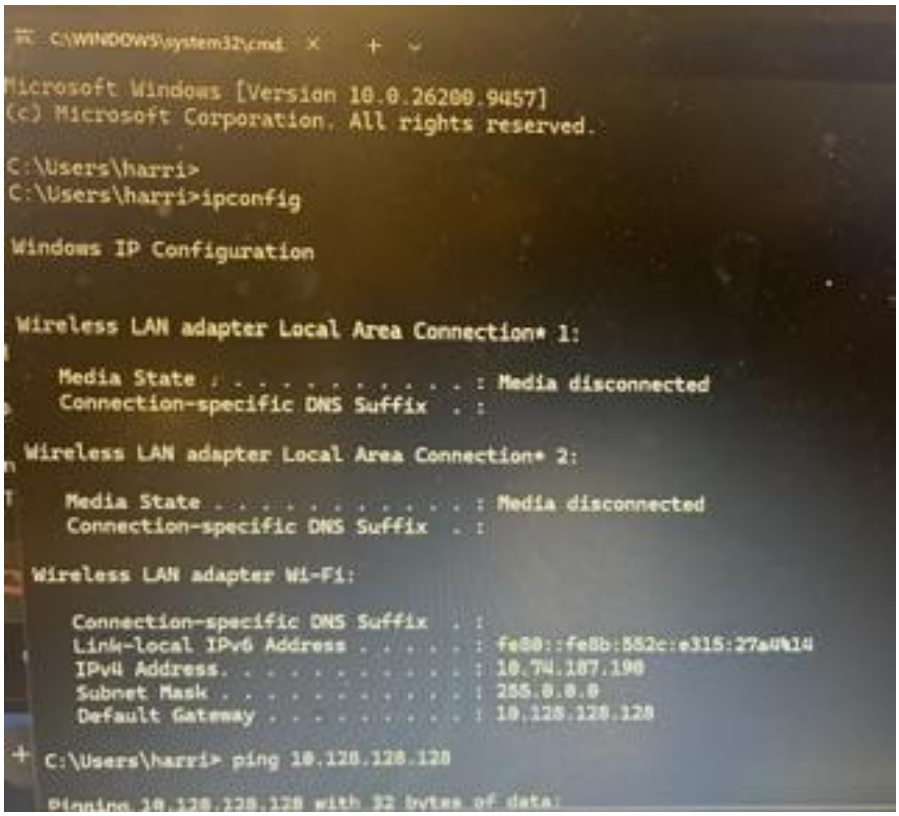
Results

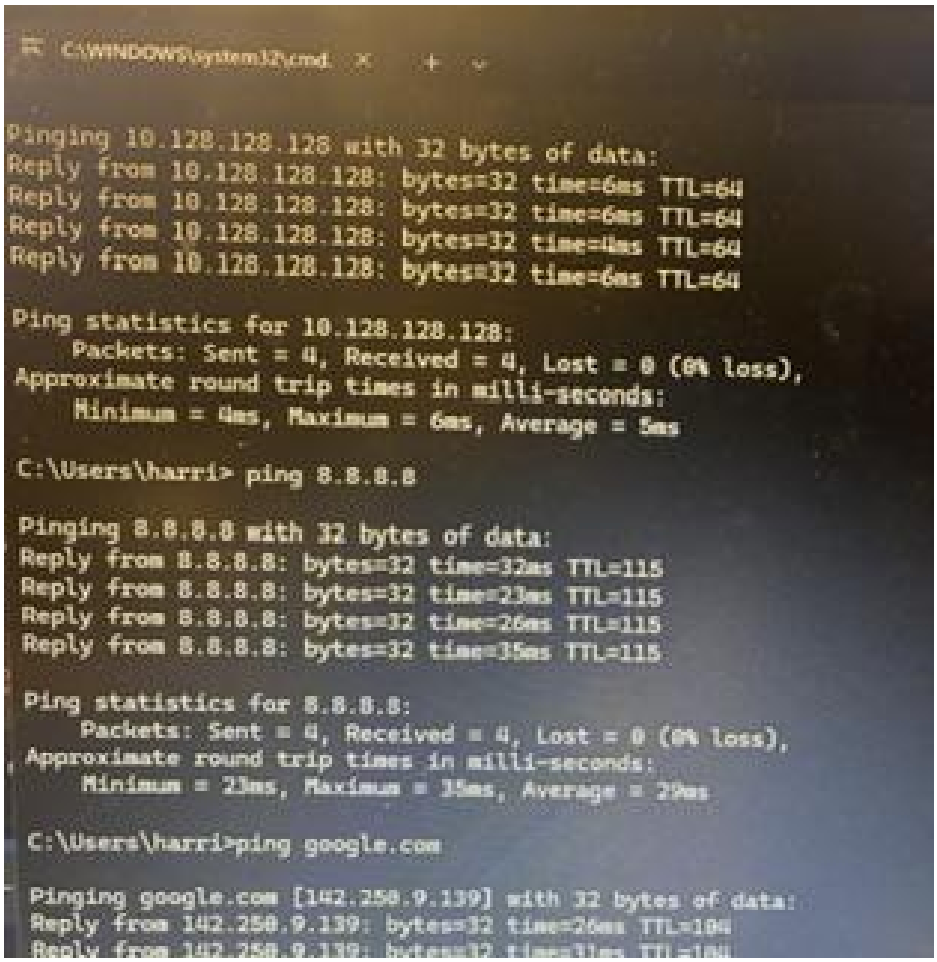
The complete troubleshooting sequence confirmed that the computer had a usable IP configuration, could communicate with the local gateway, could reach an external IP address, could resolve a domain name through DNS, and did not have a manual proxy server enabled.

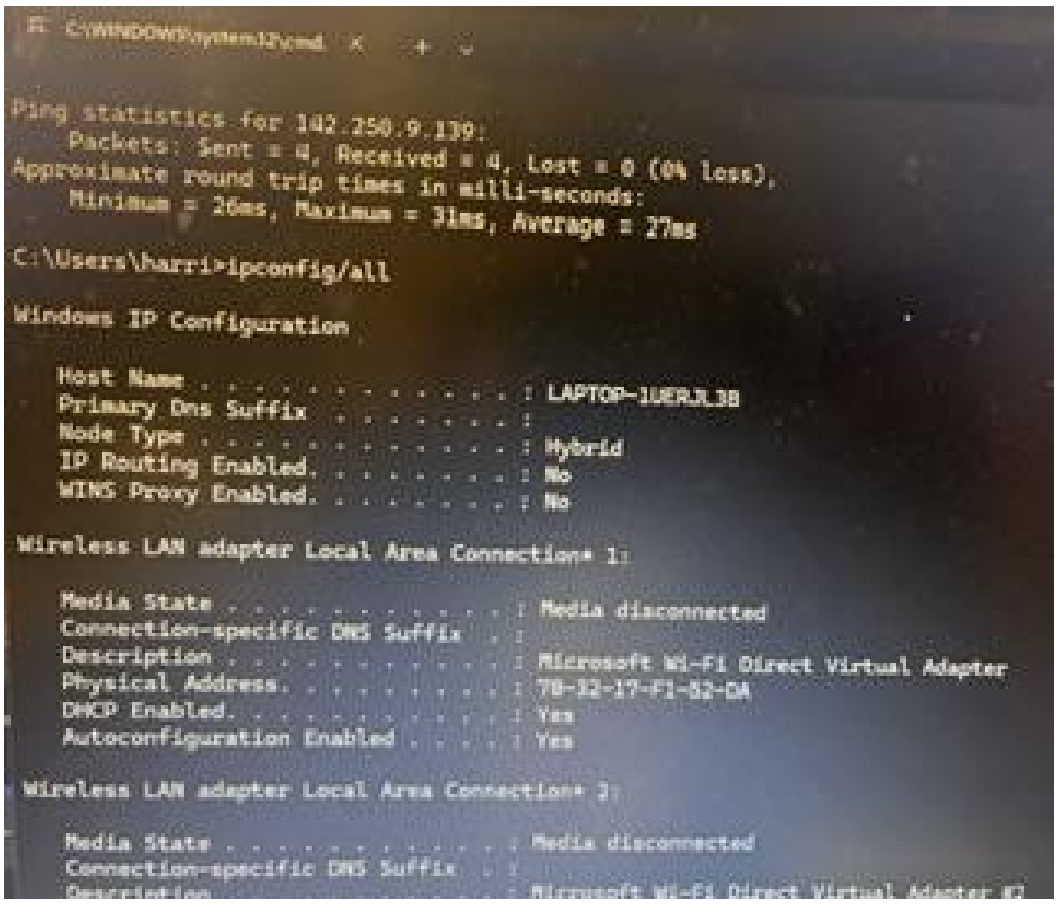
Tier 1 IT Skills Demonstrated

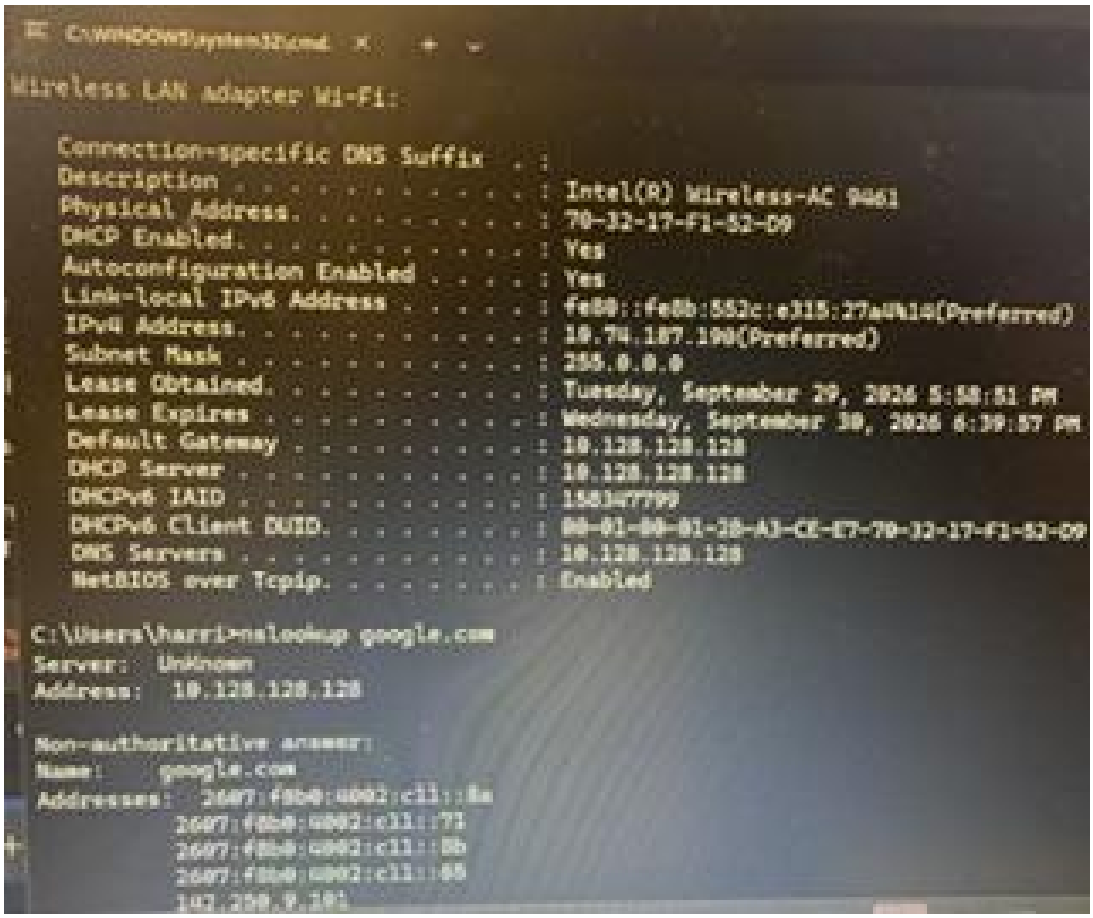
- IP configuration review
- Basic network connectivity troubleshooting
- Command-line network testing
- DNS troubleshooting and verification
- Proxy configuration verification
- Structured troubleshooting and documentation

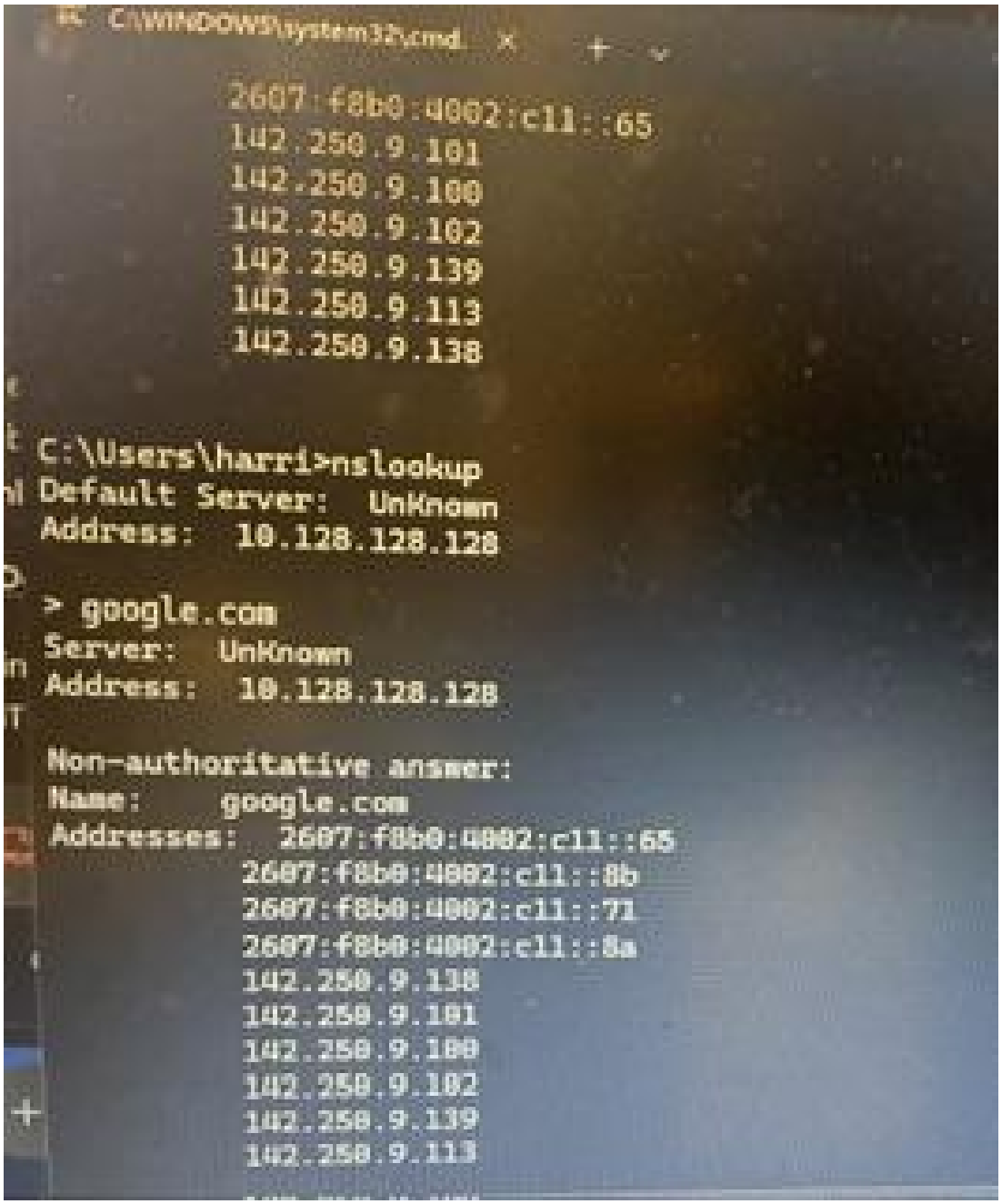
Lab Status: Completed




C:\WINDOWS\system32\cmd.
Pinging 10.128.128.128 with 32 bytes of data:
Reply from 10.128.128.128: bytes=32 time=6ms TTL=64
Reply from 10.128.128.128: bytes=32 time=6ms TTL=64
Reply from 10.128.128.128: bytes=32 time=4ms TTL=64
Reply from 10.128.128.128: bytes=32 time=6ms TTL=64
Ping statistics for 10.128.128.128:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 6ms, Average = 5ms
C:\Users\harri> ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=32ms TTL=115
Reply from 8.8.8.8: bytes=32 time=23ms TTL=115
Reply from 8.8.8.8: bytes=32 time=26ms TTL=115
Reply from 8.8.8.8: bytes=32 time=35ms TTL=115
Ping statistics for 8.8.8.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 23ms, Maximum = 35ms, Average = 29ms
C:\Users\harri>ping google.com
Pinging google.com [142.250.9.139] with 32 bytes of data:
Reply from 142.250.9.139: bytes=32 time=26ms TTL=104


C:\WINDOWS\system32\cmd.
Ping statistics for 142.250.9.139:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 26ms, Maximum = 31ms, Average = 27ms
C:\Users\harri>ipconfig/all
Windows IP Configuration
Host Name . . . . . : LAPTOP-1UERJL3B
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 78-32-17-F1-52-DA
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :


C:\WINDOWS\system32\cmd.exe
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wireless-AC 9461
Physical Address. . . . . : 70-32-17-F1-52-D9
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::fe8b:552c:e315:27a4%14(Preferred)
IPv4 Address. . . . . : 10.74.187.198(Preferred)
Subnet Mask . . . . . : 255.0.0.0
Lease Obtained. . . . . : Tuesday, September 29, 2026 5:58:51 PM
Lease Expires . . . . . : Wednesday, September 30, 2026 6:39:57 PM
Default Gateway . . . . . : 10.128.128.128
DHCP Server . . . . . : 10.128.128.128
DHCPv6 IAID . . . . . : 158347799
DHCPv6 Client DUID. . . . . : 00-01-00-01-28-A3-CE-E7-70-32-17-F1-52-D9
DNS Servers . . . . . : 10.128.128.128
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\harri>nslookup google.com
Server: UnKnown
Address: 10.128.128.128
Non-authoritative answer:
Name: google.com
Addresses: 2607:f8b0:4002:c11::8a
2607:f8b0:4002:c11::71
2607:f8b0:4002:c11::8b
2607:f8b0:4002:c11::65
142.250.9.101


C:\WINDOWS\system32\cmd.
2607:f8b0:4002:c11::65
142.250.9.101
142.250.9.100
142.250.9.102
142.250.9.139
142.250.9.113
142.250.9.138
C:\Users\harri>nslookup
Default Server: UnKnown
Address: 10.128.128.128
> google.com
Server: UnKnown
Address: 10.128.128.128
Non-authoritative answer:
Name: google.com
Addresses: 2607:f8b0:4002:c11::65
2607:f8b0:4002:c11::8b
2607:f8b0:4002:c11::71
2607:f8b0:4002:c11::8a
142.250.9.138
142.250.9.101
142.250.9.100
142.250.9.102
142.250.9.139
142.250.9.113